



AnalystLab

Africa - Data

Science Internship Programme

Week 3 Assignment

Feature Engineering Documentation

**Advanced Data Exploration & Feature Engineering
Loan Prediction Dataset**

Name: Wilson Moses

Internship: Data Science

Batch: D

Submission Date: Sun. 23.08.2026

Purpose and scope

Feature engineering was performed to improve the interpretability and modelling usefulness of the Week 2 cleaned loan dataset. The work introduced 12 new representations covering dependents, household structure, income segmentation, coapplicant contribution, repayment burden, credit interpretation and financial-variable transformations.

Design principle A useful engineered feature must have a clear business interpretation, a valid formula, a defensible modelling benefit and a documented limitation.

Starting dataset

Project measure	Result
Original cleaned observations	614
Original cleaned columns	14
Engineered features created	12
Final cleaned columns	26
Selected source modelling features	12
Final encoded modelling features	13

Data and unit considerations

The original loan_income_ratio divided a loan amount recorded in thousands by monthly income recorded in currency units. Although the calculation was internally consistent, the units were not directly comparable. The improved payment-income proxy first converts the recorded loan amount into currency units and then divides by the repayment term.

Feature catalogue: household and income

Feature	Construction	Expected value
dependents_numeric	Replace 3+ with conservative value 3	Machine-readable dependent count
family_size	1 + married indicator + dependents	Approximate household responsibility
family_size_group	Small / Medium / Large	Stakeholder-friendly segmentation
income_band	Low / Middle / High / Very High	Income-segment interpretation
has_coapplicant_income	1 if coapplicant income > 0	Single- vs dual-income signal
coapplicant_income_share	Coapplicant income / total income	Dependence on shared income

Interpretation notes

The dependents category 3+ is conservatively mapped to 3 because the precise number is unavailable. Family size is therefore an approximation rather than a verified household census. Income-band thresholds are fixed for interpretability and should be reviewed before deployment in another market or currency context.

A coapplicant income of zero is valid and does not indicate a missing value. The binary flag and income-share feature preserve that distinction while describing household-income composition more effectively than an isolated raw amount.

Household-feature selection

- o Retain dependents_numeric, has_coapplicant_income and coapplicant_income_share for modelling.
- o Keep family_size, family_size_group and income_band in the analytical dataset for reporting.
- o Avoid combining family_size with all of its exact source components when unnecessary.

Feature catalogue: credit, duration and affordability

Feature	Construction	Expected value
term_years	Loan term in months / 12	Readable repayment duration
estimated_monthly_principal	Loan amount x 1,000 / loan term	Approximate principal instalment
payment_income_ratio	Estimated monthly principal / total income	Repayment-burden proxy
credit_risk_category	Readable label from credit_history	Stakeholder-oriented risk label
log_total_income	Natural log of 1 + total income	Reduced income skewness
log_loan_amount	Natural log of 1 + loan amount	Reduced loan-amount skewness

Affordability proxy example

For the first applicant, a recorded loan amount of 128 represents approximately 128,000 currency units. Dividing by a 360-month term gives an estimated principal payment of 355.56 per month. Dividing by total income of 5,849.00 yields a payment-income proxy of 6.08%.

This calculation excludes interest, insurance, existing debt and other obligations. It is appropriate as a modelling feature, but not as a complete underwriting or affordability decision.

Redundancy controls

- o Credit-risk category duplicates credit_history; retain credit_history for modelling and the label for reports.
- o Loan term in months and term_years contain the same information; retain term_years only.
- o Raw total income and loan amount are replaced by their log-transformed versions.
- o Estimated monthly principal is represented within payment_income_ratio and is not separately retained.

Final selected modelling inputs

Feature family	Selected inputs
Applicant context	married; dependents_numeric; education; self_employed
Location and credit	property_area; credit_history
Financial transformations	log_total_income; log_loan_amount; term_years
Household and affordability	has_coapplicant_income; coapplicant_income_share; payment_income_ratio

Responsible modelling decision

Gender was intentionally excluded from predictive inputs because it can act as a protected characteristic and reproduce historical bias. It remains available in the analytical dataset for subgroup fairness auditing.

Leakage-safe preprocessing workflow

1. Split the 614 applications into 491 training rows and 123 testing rows using stratified sampling.
2. Fit one-hot encoding on training categories only and ignore unseen categories during test transformation.
3. Fit standard scaling on the training values of continuous engineered features only.
4. Pass binary and ordinal variables through without unnecessary scaling.
5. Export cleaned, combined ML-ready, training and testing datasets with an updated data dictionary.

Conclusion

The final feature set combines the strongest statistical evidence with interpretable financial relationships, controlled redundancy, fairness considerations and reproducible train/test preprocessing. These decisions prepare the project for reliable Week 4 model development.